

## Using SAS to Estimate Variance Components in One Way Random Effects Model

We have seen in Lecture Notes no. 6, how with the help of PROC GLM one can obtain point estimates for the variance components in a one way random effects model. We also have seen in Lecture notes no 6 how one can construct confidence intervals for the variance components in a one way random effects model, and also a confidence interval for the proportion of variability due to the variance component of the random factor. We have seen that this last interval was not informative as it covered a wide range with the interval (0,1).

PROC GLM in SAS cannot provide estimates for variance components in a one way random effects model, nor the confidence intervals. The following SAS procedure, PROC VARCOMP, can accomplish this.

For the data set that was analyzed for a one way random effects model, Personnel Officers, the following SAS program will give use in addition to the results we obtained before, the point estimates of the variance components and their 95% confidence intervals.

```
TITLE ' PERSONNEL OFFICERS SCORING STUDY';
DATA APEX;
INPUT OFFICER SCORE @@;
CARDS;
1 76 1 64 1 85 1 75
2 58 2 75 2 81 2 66
3 49 3 63 3 62 3 46
4 74 4 71 4 85 4 90
5 66 5 74 5 81 5 79
PROC GLM PLOTS=(DIAGNOSTICS);
CLASS OFFICER;
MODEL SCORE=OFFICER;
OUTPUT OUT=NEW R=RESID;
PROC UNIVARIATE DATA=NEW NORMAL;
VAR RESID;
PROC VARCOMP DATA=APEX METHOD=TYPE1;
CLASS OFFICER;
MODEL SCORE=OFFICER/CL;
RUN;
```

The SAS output file: “SAS file for personnel officers with varcomp.pdf” presents the results we obtained before, the point estimates of the variance components and their 95% confidence intervals. These numerical results are identical for the point estimates, and very close to the ones for the two confidence intervals, that were presented in Lecture Notes no 6.

Note that we do not get from this PROC VARCOMP the point estimate or the confidence interval for the proportion of variability due to the random effect.

Remark. It also works for a non balanced one way ANOVA!